

PALTINIS SIBIU, Romania

WMO: 152540

Lat: 45.6572N

Lon: 23.9325E

Elev: 4771

StdP: 12.33

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.4	8.3	-5.4	4.9	12.7	0.4	6.7	17.4	35.2	34.5	20.8	26.2	2.7	60	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	75.0	61.4	71.9	59.6	69.2	58.3	62.9	71.2	61.3	68.8	60.1	67.2	3.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.1	92.7	65.3	58.6	87.8	64.2	57.3	83.6	63.5	31.0	70.8	29.8	69.1	28.8	67.2	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	17.4	12.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	25.1	28.9	31.2	39.9	48.7	55.8	57.5	59.3	51.4	44.2	37.9	28.6
	DBStd	14.28	9.62	8.64	8.17	10.07	6.84	6.84	5.80	6.16	8.86	8.73	8.51	8.82
	HDD50	3758	772	590	583	325	103	25	8	6	92	223	367	664
	HDD65	8268	1237	1010	1047	753	504	283	240	193	414	646	812	1129
	CDD50	1001	0	0	1	22	64	199	241	294	134	42	4	0
	CDD65	36	0	0	0	0	0	6	8	16	6	0	0	0
	CDH74	134	0	0	0	0	0	18	36	64	16	0	0	0
	CDH80	7	0	0	0	0	0	1	1	3	2	0	0	0
(14) Wind	WSAvg	3.1	2.9	3.7	3.7	3.8	3.3	2.5	2.3	2.2	3.0	2.5	3.7	3.6
(15) Precipitation	PrecAvg	31.5	1.3	1.2	1.7	2.9	3.9	4.5	4.7	3.2	2.9	2.1	1.7	1.7
	PrecMax	41.1	3.0	3.2	3.3	5.5	7.4	7.7	8.3	5.9	6.2	4.9	5.0	4.1
	PrecMin	20.9	0.4	0.4	0.7	0.7	1.5	1.1	1.6	0.6	0.5	0.0	0.1	0.4
	PrecStd	5.8	0.7	0.6	0.7	1.2	1.7	1.8	1.9	1.3	1.5	1.1	1.0	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	50.8	56.1	66.3	69.1	76.2	77.9	79.0	75.7	67.3	59.9	51.0
		MCWB	37.9	39.6	43.7	50.8	54.7	60.8	61.6	62.8	59.6	51.7	45.9	36.8
	2%	DB	42.6	46.6	49.8	61.4	65.4	72.6	73.9	75.5	70.6	63.4	54.7	45.3
		MCWB	35.8	39.0	39.7	48.7	53.5	59.7	61.4	61.4	56.8	50.9	44.0	36.9
	5%	DB	39.5	43.0	46.1	57.0	62.5	69.6	69.9	71.6	66.5	59.1	51.3	41.9
		MCWB	34.0	36.1	37.7	46.3	52.4	58.9	59.5	59.9	55.2	49.1	42.8	36.0
	10%	DB	36.9	39.7	42.3	53.1	59.2	66.5	66.6	68.7	63.1	55.3	48.5	39.1
		MCWB	33.0	34.7	35.9	44.4	51.1	57.7	58.4	58.8	54.4	47.2	41.7	34.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	43.2	44.0	52.8	57.5	64.3	64.5	65.2	61.1	55.2	47.5	41.6
		MCDB	44.4	47.8	54.4	62.8	66.1	70.7	72.9	75.3	71.0	62.2	53.9	45.7
	2%	WB	37.0	39.6	41.1	49.8	54.9	61.9	62.5	63.1	58.5	52.5	45.9	38.7
		MCDB	40.8	46.2	49.3	60.1	62.1	69.4	69.9	72.7	67.8	60.0	51.3	43.3
	5%	WB	35.3	37.2	38.7	47.5	53.3	60.2	61.0	61.2	56.5	50.5	44.8	36.9
		MCDB	38.9	41.7	44.9	55.9	60.7	67.9	67.2	69.3	64.1	57.6	49.5	40.6
	10%	WB	33.5	35.3	36.6	45.3	51.6	58.6	59.7	59.9	55.0	48.7	43.0	35.2
		MCDB	36.3	39.4	41.4	52.0	58.1	65.8	65.6	67.4	61.9	54.8	47.3	38.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.4	12.4	13.2	12.9	14.1	14.3	14.0	12.3	12.7	10.9	10.6
		MCDBR	14.6	15.6	17.1	18.0	16.6	17.7	17.7	17.0	16.0	16.7	14.1	13.8
		MCWBR	9.8	10.2	10.6	11.1	10.0	10.2	11.1	10.2	10.0	10.9	9.2	8.8
	5% WB	MCDBR	12.8	13.6	15.4	17.4	15.5	16.8	15.7	15.7	14.9	15.5	12.1	12.0
		MCWBR	9.0	8.8	9.8	10.9	10.1	10.6	10.6	10.3	9.6	10.6	8.0	8.4
(40) Clear-Sky Solar Irradiance	taub		0.264	0.288	0.323	0.372	0.361	0.359	0.373	0.382	0.345	0.314	0.292	0.266
	taud		2.409	2.359	2.348	2.265	2.344	2.371	2.329	2.300	2.391	2.484	2.496	2.447
	Ebn at Noon		272	284	288	281	287	288	282	274	276	272	259	259
	Edh at Noon		22	29	34	40	38	37	39	38	32	25	21	19
(44) All-Sky Solar Radiation	RadAvg		424	680	1026	1366	1642	1814	1809	1648	1186	827	486	345
	RadStd		49	85	129	150	156	169	155	154	122	100	56	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (77 neighbors)	+1.21	N/A	N/A	+1.62	N/A	N/A	-221	-329	+166	+65

Nomenclature: See separate page